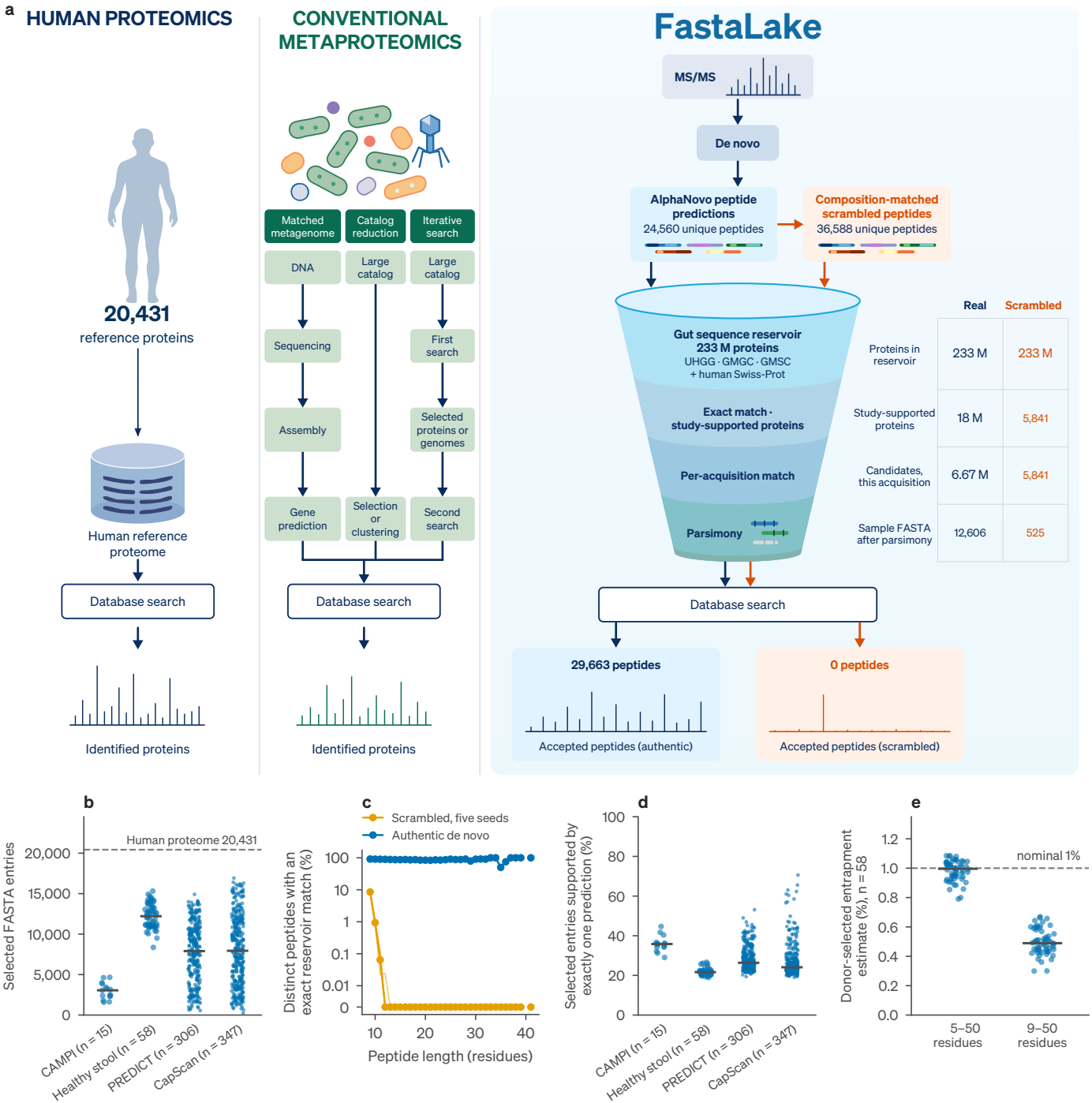

a
HUMAN PROTEOMICS
20,431
reference proteins
Human reference proteome
Database search
Identified proteins
CONVENTIONAL METAPROTEOMICS
Matched metagenome
Catalog reduction
Iterative search
DNA
Large catalog
Large catalog
Sequencing
First search
Assembly
Selected proteins or genomes
Gene prediction
Selection or clustering
Second search
Database search
Identified proteins
FastaLake
MS/MS
De novo
AlphaNovo peptide predictions
24,560 unique peptides
Composition-matched scrambled peptides
36,588 unique peptides
Gut sequence reservoir
233 M proteins
UHGG · GMGC · GMSC
+ human Swiss-Prot
Exact match · study-supported proteins
Per-acquisition match
Parsimony
Real
Scrambled
Proteins in reservoir
233 M
233 M
Study-supported proteins
18 M
5,841
Candidates, this acquisition
6.67 M
5,841
Sample FASTA after parsimony
12,606
525
Database search
29,663 peptides
Accepted peptides (authentic)
0 peptides
Accepted peptides (scrambled)
b
Selected FASTA entries
Human proteome 20,431
20,000
15,000
10,000
5,000
0
CAMPI (n = 15)
Healthy stool (n = 58)
PREDICT (n = 306)
CapScan (n = 347)
c
Scrambled, five seeds
Authentic de novo
Distinct peptides with an exact reservoir match (%)
100
10
1
0.1
0.01
0
10
20
30
40
Peptide length (residues)
d
Selected entries supported by exactly one prediction (%)
100
80
60
40
20
0
CAMPI (n = 15)
Healthy stool (n = 58)
PREDICT (n = 306)
CapScan (n = 347)
e
Donor-selected entrapment estimate (%), n = 58
1.2
1.0
0.8
0.6
0.4
0.2
0.0
nominal 1%
5–50 residues
9–50 residues